Subtitle Basic idea about DC to AC converter
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Subject Power Electronics EE 603

Inverter Lecture 06

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Hello friends

Lecture 06

1. Brief recap of last lecture 05
2. Sinusoidal pulse width modulation Inverter
3. Signification
4. References

PWM Inverter

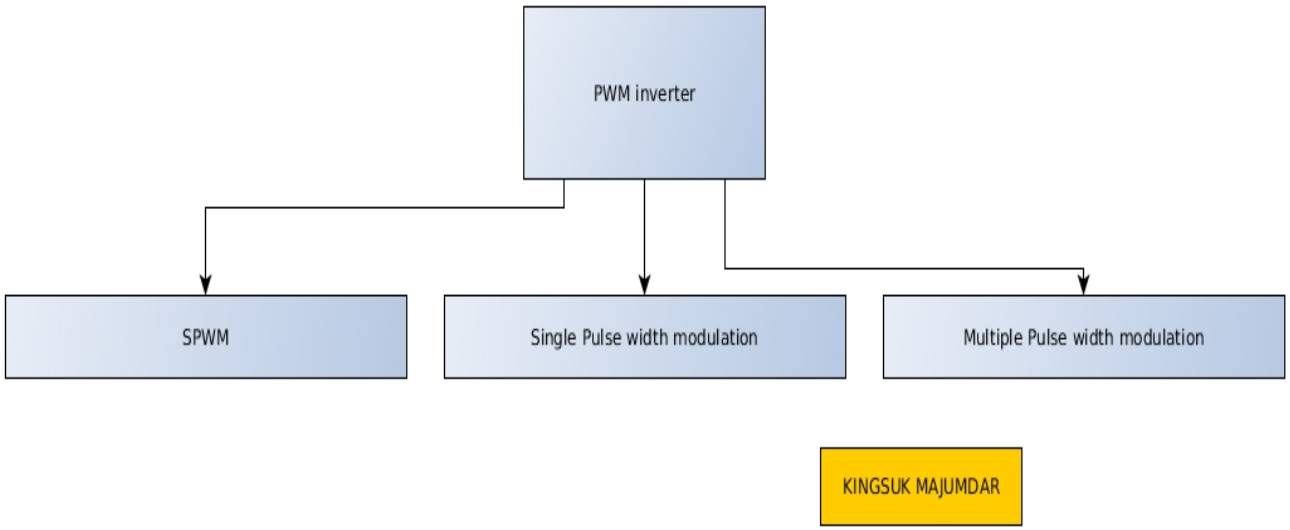


Figure: PWM inverter type

Sinusoidal pulse width modulation Inverter

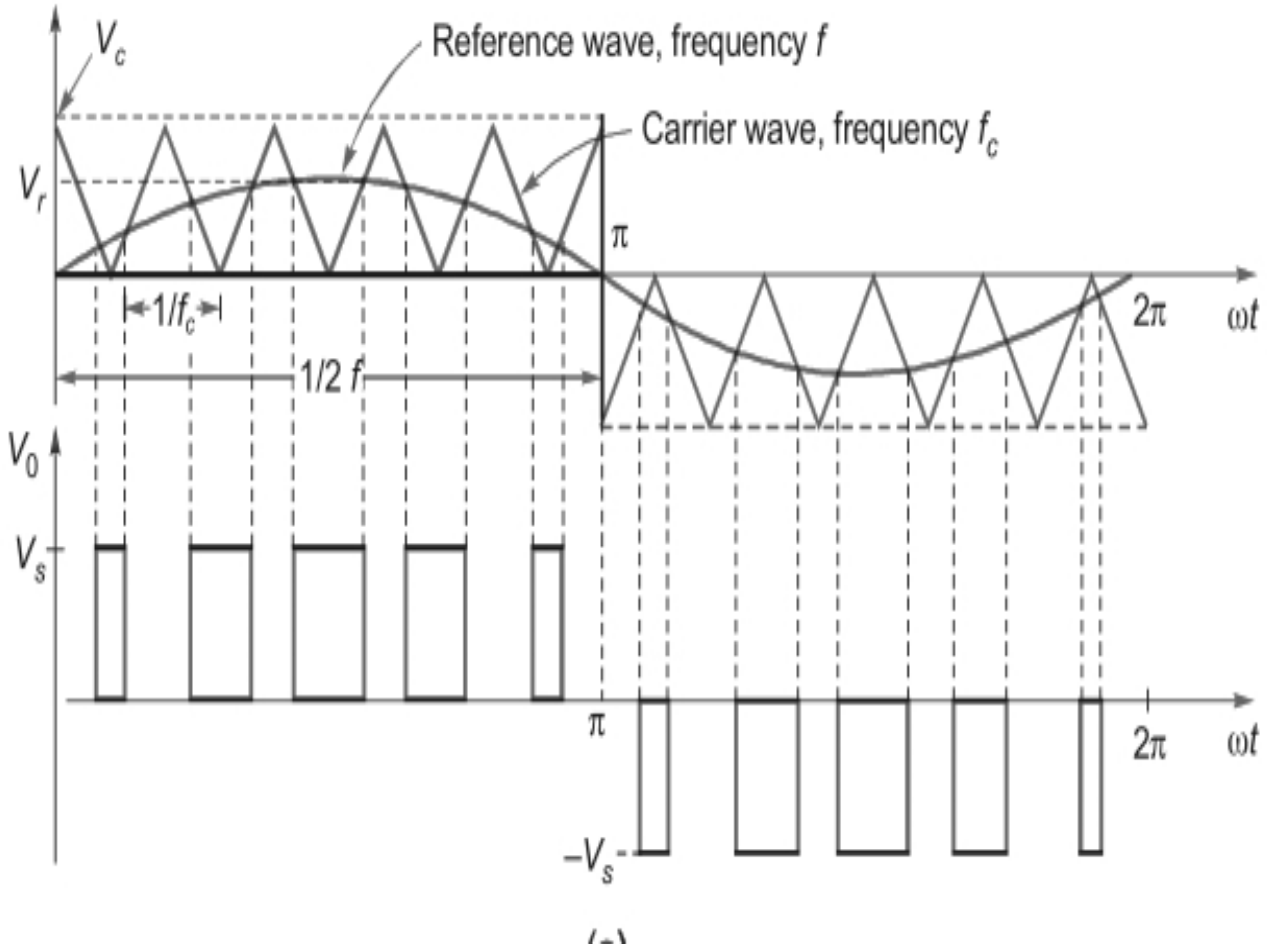


Figure: Multiple pulse width modulation Inverter

NB:picture taken from google

AM index and FM index

- AM index (Amplitude modulation index)

$$M_a = \frac{A_{ref}}{A_{car}} \tag{1}$$

- FM index (Frequency modulation index)

$$M_f = \frac{f_{car}}{f_{out}} \tag{2}$$

- N,no of pulse per half cycle
  another name: Uniform pulse width modulation Inverter

$$N = \frac{f_{car}}{2 * f_{out}} = \frac{M_f}{2} \tag{3}$$

- The instantaneous output voltage

$$v_o = V_s(g_1 - g_4) \tag{4}$$

Circuit for SPWM Inverter

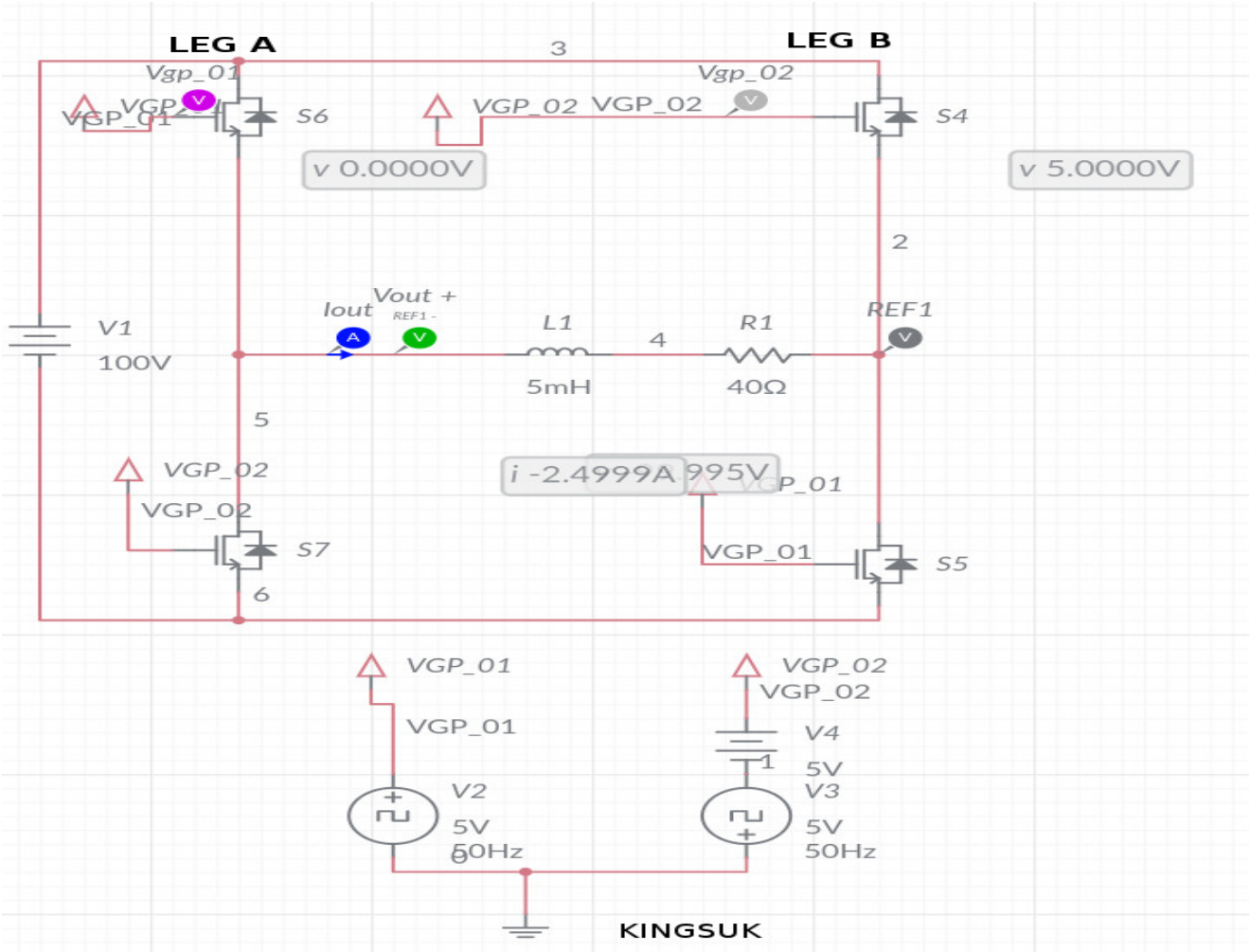


Figure: Circuit for SPWM inverter

SPWM explanation

The dc to ac converter i.e. inverter has an internal control as PWM. The sinusoidal PWM one of them, in where the pulse width is a sinusoidal function of the angular position of the pulse in a cycle.

- Carrier wave (high frequency wave): Triangular
- Reference wave (low frequency): Sinusoidal

Number of pulse per half cycle can be written as follows

$$N = \begin{cases} \frac{f_c}{2 * f_r}; \text{for case a} \\ \frac{f_c}{2 * f_r} - 1; \text{for case b} \end{cases} \tag{5}$$

- Case a: If the peak of carrier coincident with zero of reference (see fig: 2)
- Case b: If the zero of carrier coincident with zero of reference (see Fig 4)

cont...

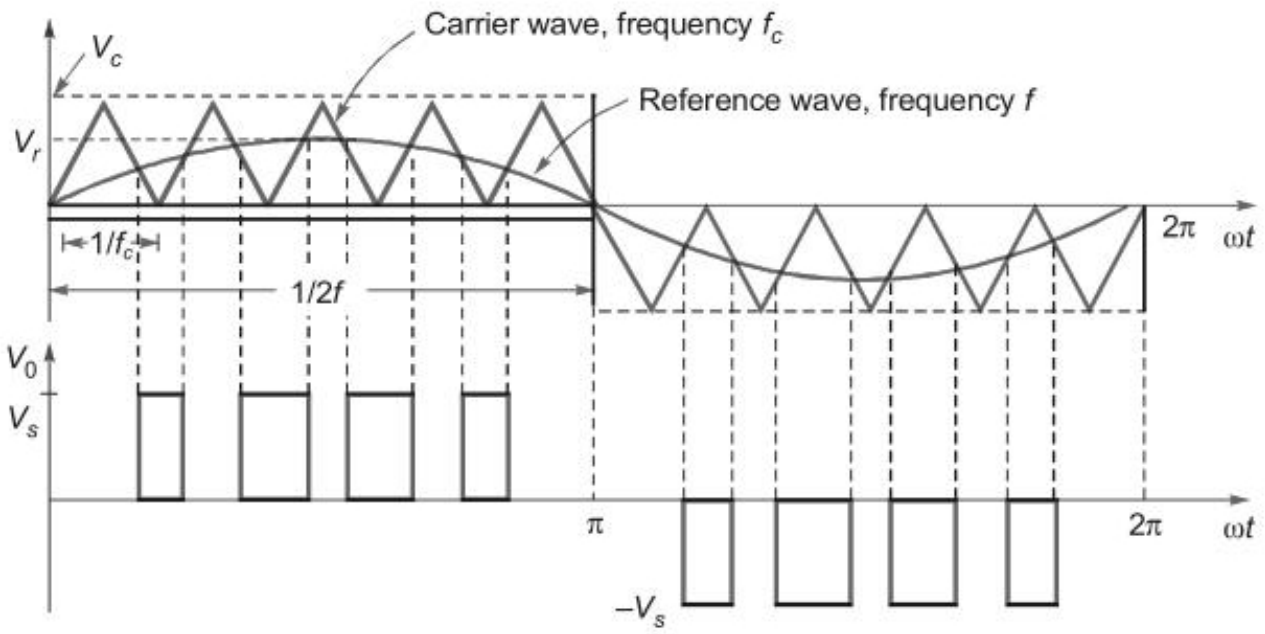


Figure: SPWM inverter (case B)

Signification:

The M_a can be two type in the case of SPWM

- Linear Modulation ($M_a \leq 1$): The fundamental-frequency component in the output voltage varies linearly with the amplitude of the modulation index M_a
- Over Modulation ($M_a > 1$): The peak of the reference voltage is allowed to exceed the triangular voltage.

cont...

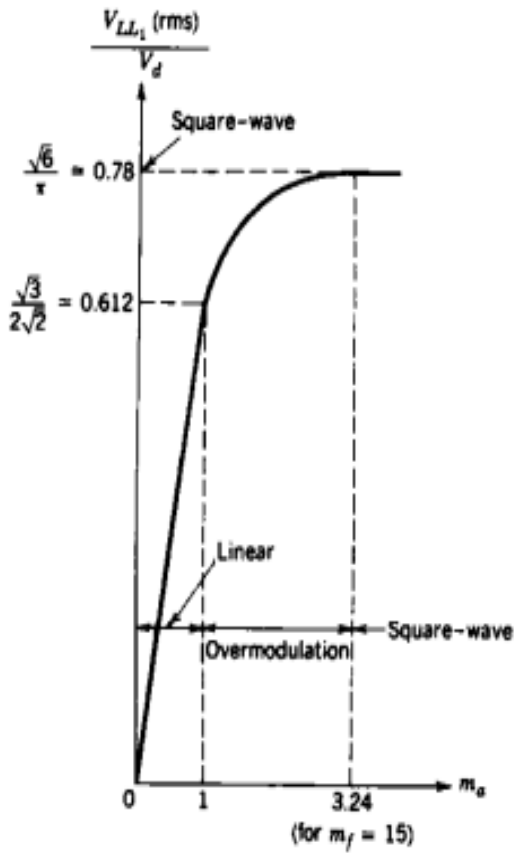


Figure: M_a variation in SPWM

taken from book[2]

References


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